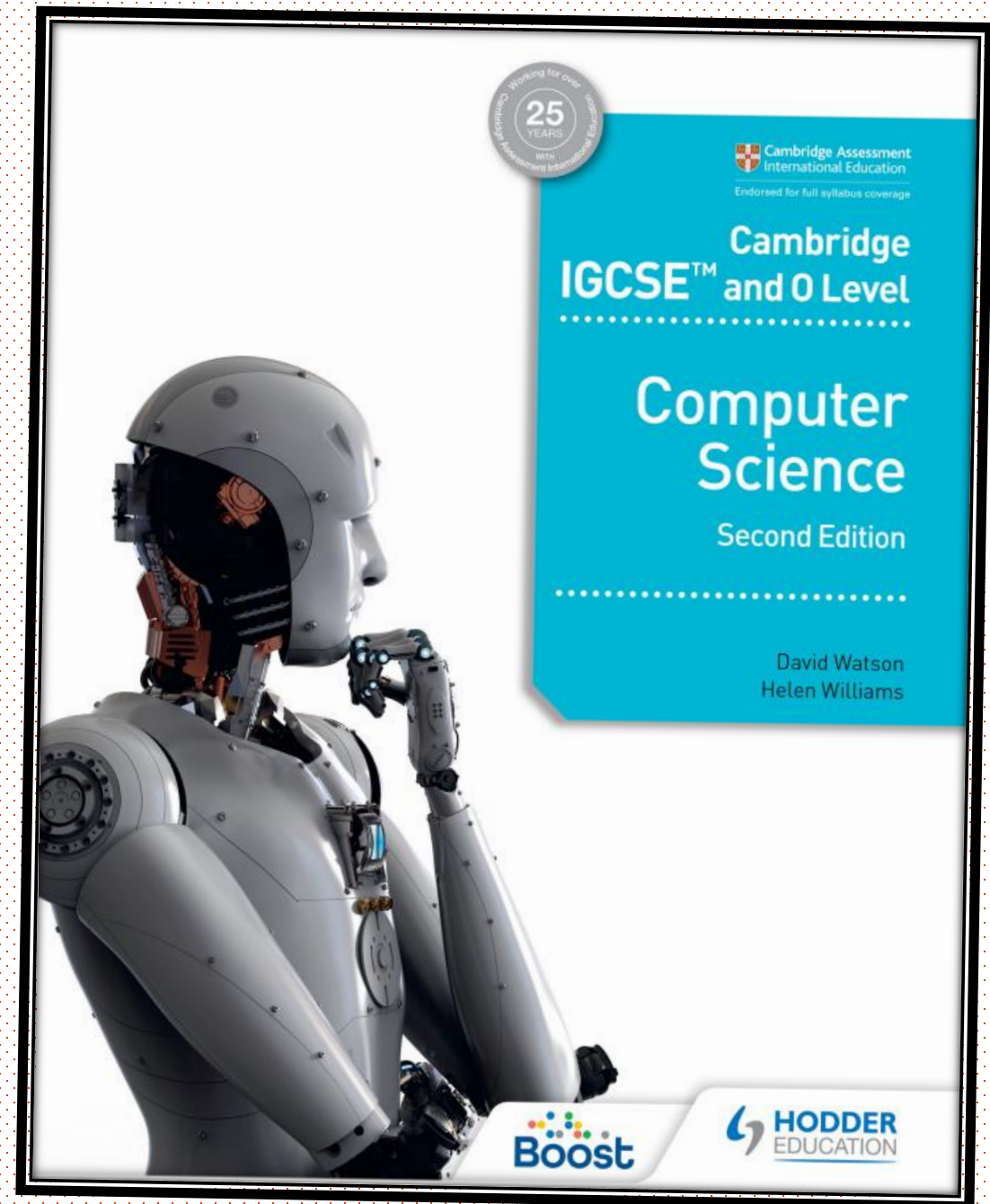


SYLLABUS CODES (0478,2210)

DATA

LEC-14



WHAT IS DATA?

Data	By Itself
25	Is it age? Temperature? Score? We don't know
A	Is it a grade? First letter of name? We don't know
Cat	Just three letters — could mean the animal or something else

Data is raw facts and figures that have no meaning by themselves.

Data	Information
25°C (just a number)	"It is 25°C outside — I should wear shorts"
Raw facts	Data given meaning

- Data + Meaning = Information.
- Data is raw facts that a computer can store, process, or transmit.



DATA VS INFORMATION

Type of Data	Stored As
Text	ASCII / Unicode codes
Image	Pixels (binary values for colors)
Audio	Sound samples (amplitude values)
Video	Many images + audio track



DATA: COMPUTER SCIENTIST VS LAYMAN

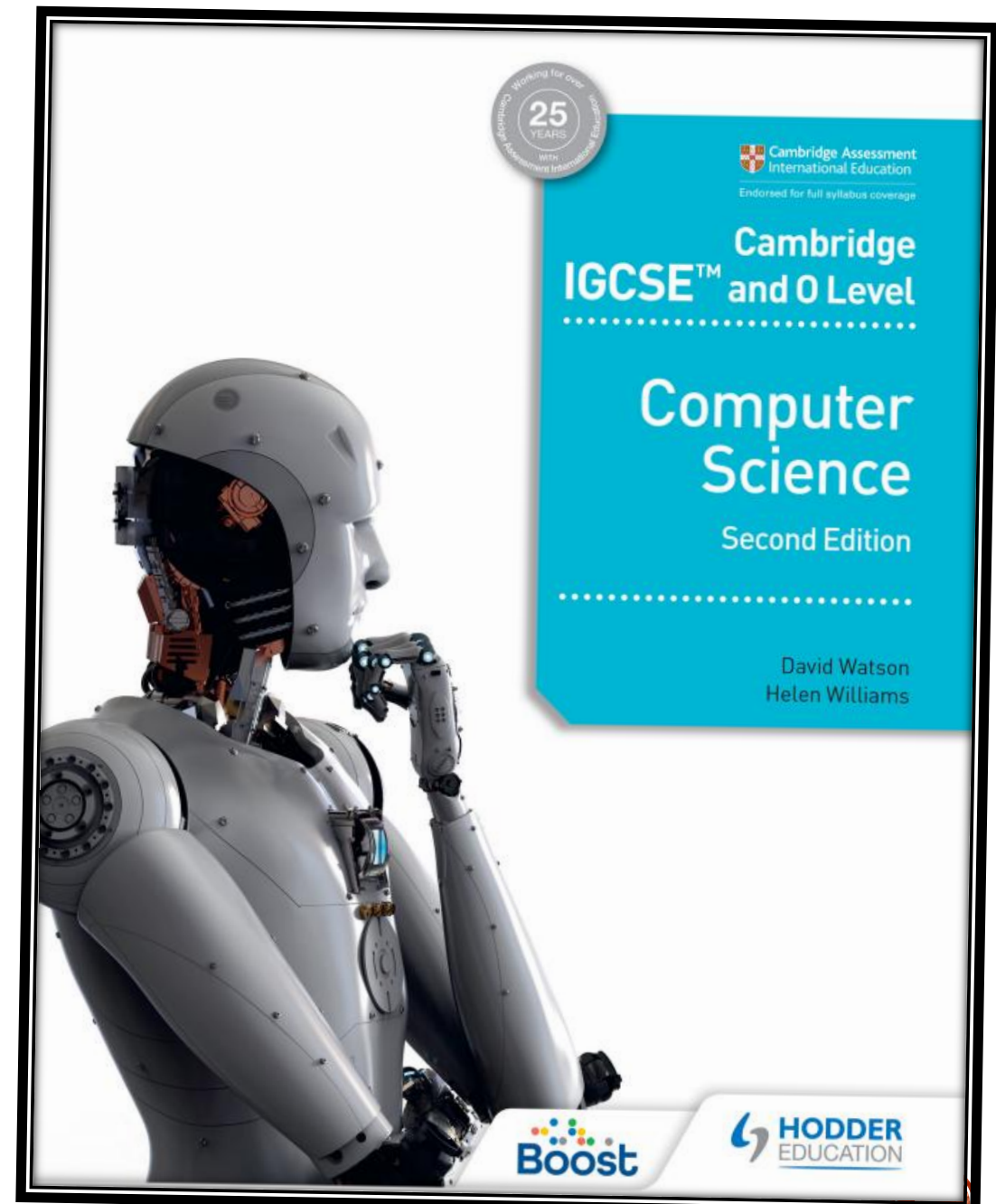
Aspect	Layman (Ordinary Person)	Computer Scientist
What is data?	Information like text, photos, songs, videos	Raw binary values (0s and 1s) stored in memory
Example of data	"A beautiful sunset photo"	11111111 00000000 00000000 00000000 (pixel values)
Where data lives	On phone, computer, USB drive	In registers, RAM, hard disk, cache
How they think of size	"This photo takes too much space"	"This image is 4 MB = 4,000,000 bytes"
Data types	Documents, pictures, music, movies	Text (ASCII/Unicode), Integer, Float, Boolean, Array, etc.
What happens to data	Open, edit, save, share, delete	Read, write, copy, move, encode, decode, compress
Data problem	"I lost my files!"	"Memory leak, buffer overflow, data corruption"



SYLLABUS CODES (0478,2210)

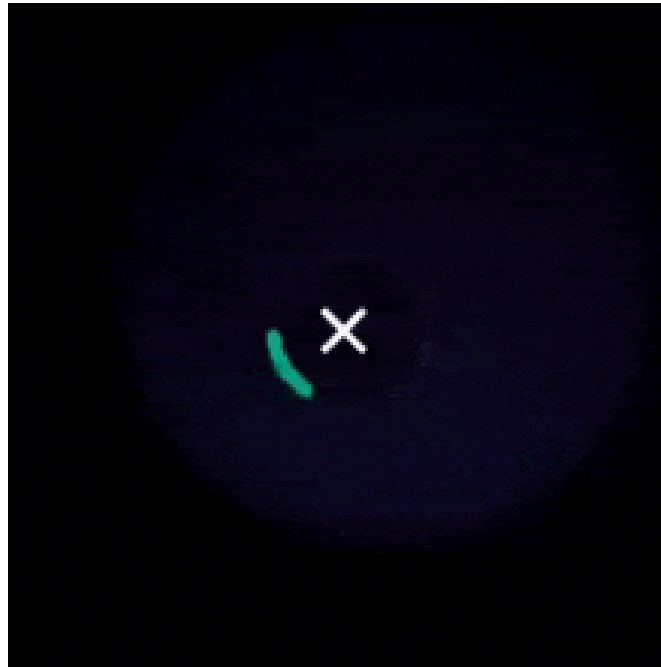
DATA PACKETS

LEC-15



DATA PACKETS

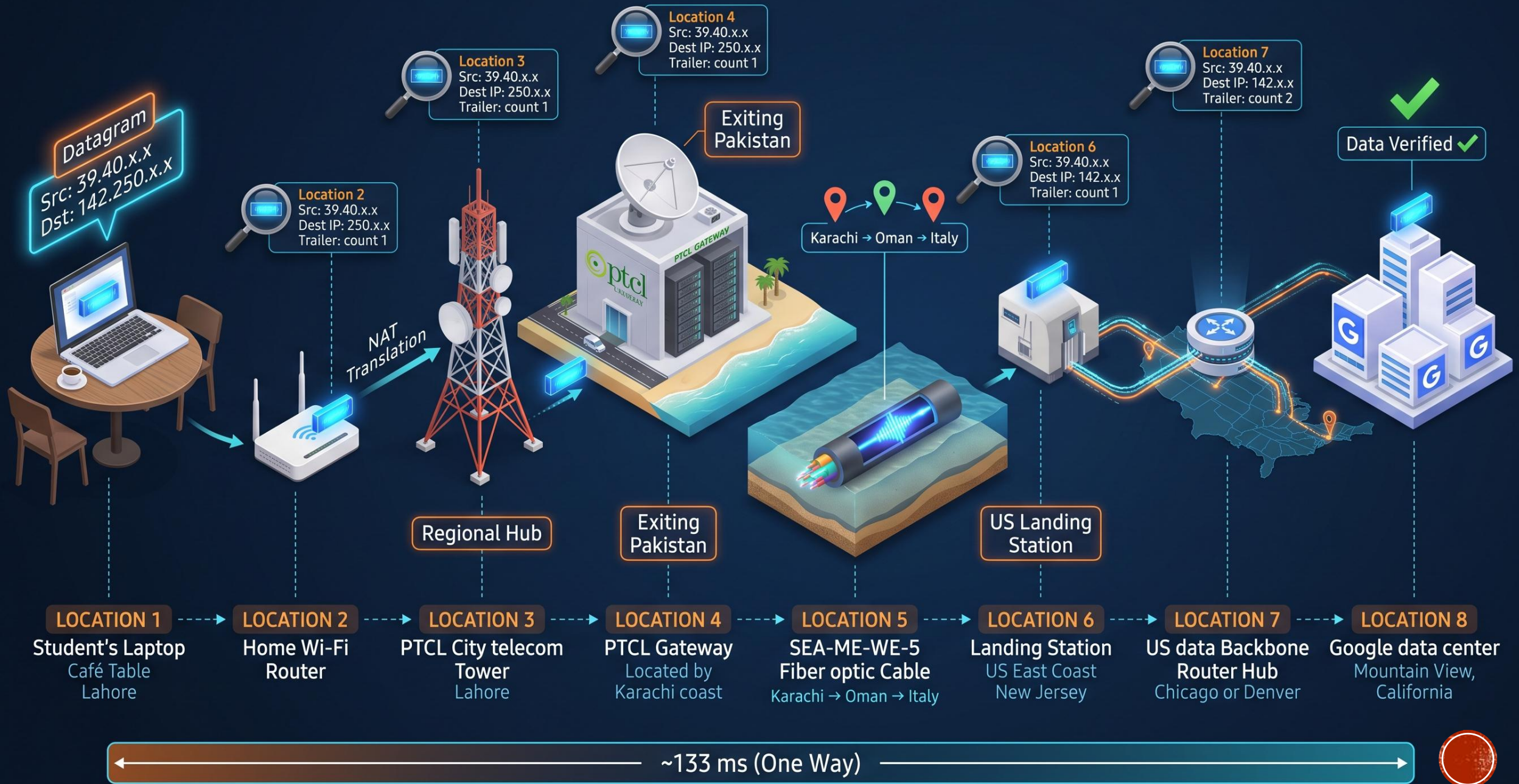
- What happens you send an image to a friend or family member on WhatsApp?
- Is it received at once or sometimes takes time to download and shows loading...?



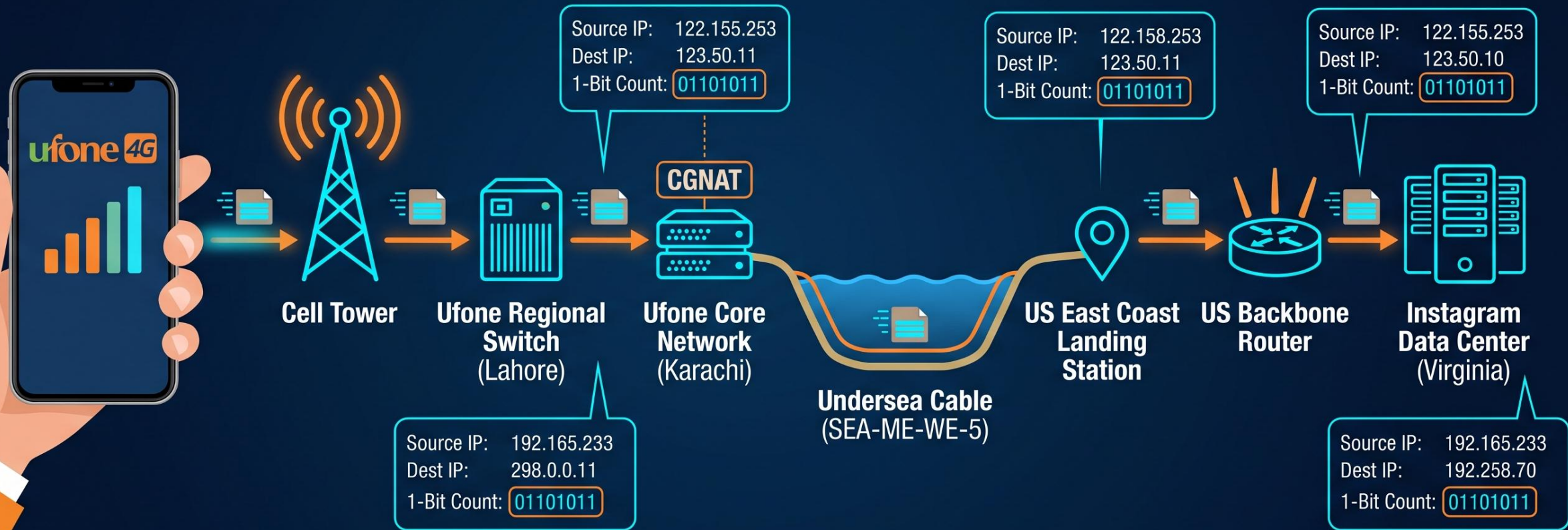
DATA PACKETS

- Data sent over long distance is broken up into chunks (called datagrams/packets).
- The packets of data are usually quite small, typically 64KiB.
- The data packets are sent along a different route to its destination.
- The only drawback of splitting data into packets is that it needs to reassemble the data when it reaches its destination.





Data Packet from Lahore, Pakistan) to Instagram Servers (Virginia, USA)

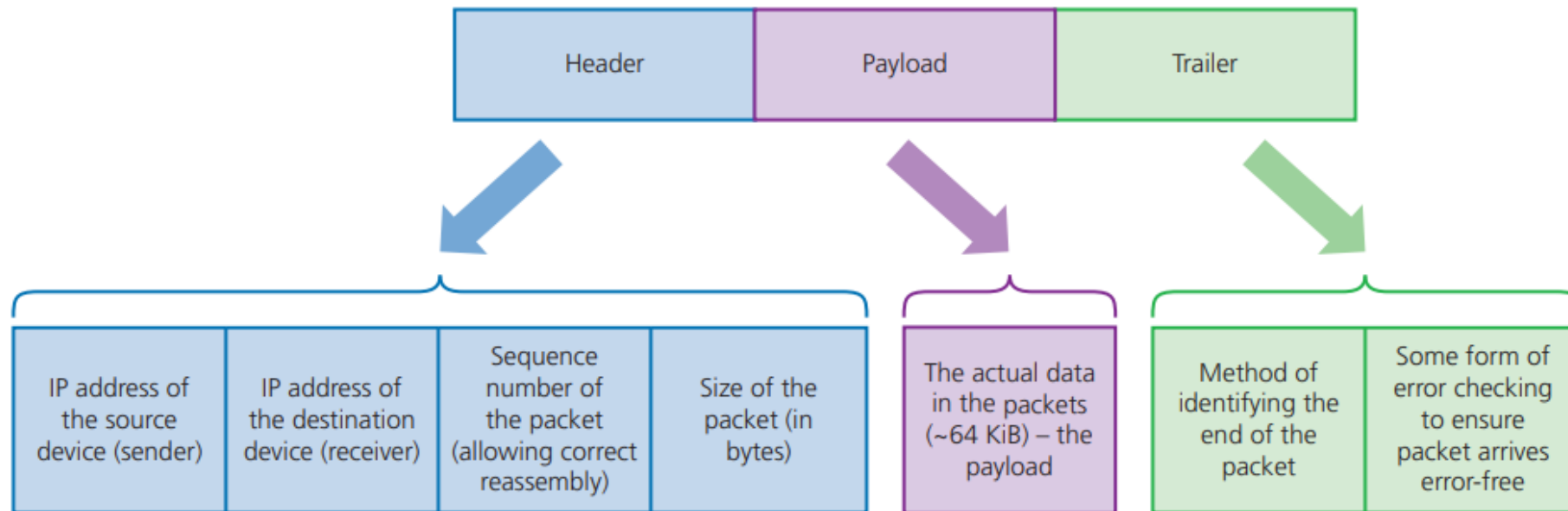


~111 ms One Way



PACKET STRUCTURE

- A typical packet is split up into:
 - a packet header
 - the payload
 - a trailer



PACKET STRUCTURE CONT. . . .

Header	Payload	Trailer
the IP address of the sending device	The actual data in the packets (~64 KiB) – the payload	End marker: Identifies where the packet ends
the IP address of the receiving device		
Sequence number of the packet (allowing correct reassembly)		Error checking (CRC = Cyclic Redundancy Check): Checks if data was corrupted during transmission
Size of the packet (in bytes)		



CYCLIC REDUNDANCY CHECK - SIMPLIFIED

1. Before sending, the sender counts the 1-bits in the data and stores that count in the packet's trailer.
2. The receiver gets the packet, counts the 1-bits in the data again.
3. The receiver compares its count with the count in the trailer.
4. If they match → no errors. If they don't match → ask to resend.
5. *Example: The letter "A" is stored as the binary number 01000001 (2s)*

